Arpit Jaipal Makheja

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.38
Nirala junior college	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education)	2024	91.33%
The Royal Gondwana Public School	Class X (Central Board of Secondary Education)	2022	95.00%

SKILLS

Programming Languages and Tools: C, C++, Python, R, SQL, Git, Github, Google Colab, Jupyter Notebook, VS Code, Terminal

AI/ML: Machine Learning, Feature Engineering, Model Evaluation, Time-Series Analysis, RAG, Vector Databases, Embeddings

Libraries and Frameworks: Pandas, NumPy, Matplotlib, Scikit-learn, Tensorflow, Langchain, Langraph

Computer Science: Data Structures & Algorithms, Object-Oriented Programming, DBMS, REST APIs, Problem Solving

Soft Skills: Leadership quality, Public Speaking, Team Collaborations, Event Management, Corporate Outreach

WORK EXPERIENCE

Intern- CapturAI

- Engineered an AI-readiness evaluation system for user-submitted webpages by analyzing **content and structured-data signals**, enabling prediction of LLM retrieval likelihood.
- Developed a weighted scoring engine combining **20+ webpage quality parameters** to generate static quality, AI-readiness, and structured-data scores for content evaluation.
- Designed a system to analyze webpage content, metadata, and HTML structure, generating **AI-readiness dashboards** with actionable insights into LLM discoverability and content quality.
- **Automated** the LLM provider verification testing pipeline, **reducing false positives by 34%** through improved cross-provider validation logic.

PROJECTS

WebRAG : AI-Powered Website RAG Assistant

- Developed and deployed an end-to-end Website RAG application that enables natural-language querying over **entire websites** by recursively crawling, indexing, and semantically retrieving webpage content.
- Designed a modular RAG architecture separating **ingestion, retrieval, generation, and evaluation** layers, enabling seamless integration of multiple embedding models and LLM backends without modifying core pipeline logic.
- Implemented a complete retrieval pipeline using **LangChain, ChromaDB, BAAI/bge-base-en-v1.5 embeddings, and Gemini 2.5 Flash**, combining recursive crawling, semantic chunking, vector search, and answer generation.

Finance Analytics & Recommender System

- Designed a stock analytics platform in Python and Streamlit, covering **150+ NIFTY-listed stocks** with live market data, **candlestick pattern detection**, and technical indicators feeding a **next-day price movement** classifier.

- Trained **Random Forest** and **XGBoost** models using a time-based train/test split to avoid data leakage, and built a **K-Means and cosine-similarity** recommender system to surface comparable stocks within an interactive dashboard.

ML-Based Predictive Maintenance Model

- Built a predictive maintenance pipeline for industrial sensor data by performing **feature engineering, exploratory analysis, categorical encoding**, and class-imbalance handling on machine failure dataset.
- Trained and tuned Logistic Regression, SVM, and XGBoost classifiers using **GridSearchCV**, with **XGBoost** achieving the best results (88% overall accuracy and 0.78 recall on the minority failure class).

Inventory Management System

- Built a C++ based inventory management system with persistent file storage, enabling accurate real-time tracking of **product quantity, price, and category** information consistently across multiple user sessions.
- Implemented robust and modular backend logic in C++, effectively utilizing core Object-Oriented Programming principles such as **abstraction, inheritance, and polymorphism** to structure the overall system.

POSITIONS OF RESPONSIBLITY

Core at Tinkerer's Lab IIT Hyderabad:

- Led outreach to 100+ companies, pitching collaboration and sponsorship opportunities to support Thrust 2026.

Coordinator at Finance and Consulting Club IIT Hyderabad:

- Collaborated with a 20-member ideation team at FCC IITH, driving event strategy and planning for Horizon 2026

RELEVANT COURSES

AI/ML: Introduction to Artificial Intelligence, Machine Learning for Bioinformatics, Biostatistics, Big Data Biology

Computer Science: Data Structure and Applications , Introduction to programming , R and python for biologists

PROGRAMMING ACHIEVEMENTS

Solved **400+ algorithmic problems** across LeetCode and GeeksforGeeks, strengthening proficiency in data structures, algorithms, and competitive problem solving while earning **2 platform badges**.